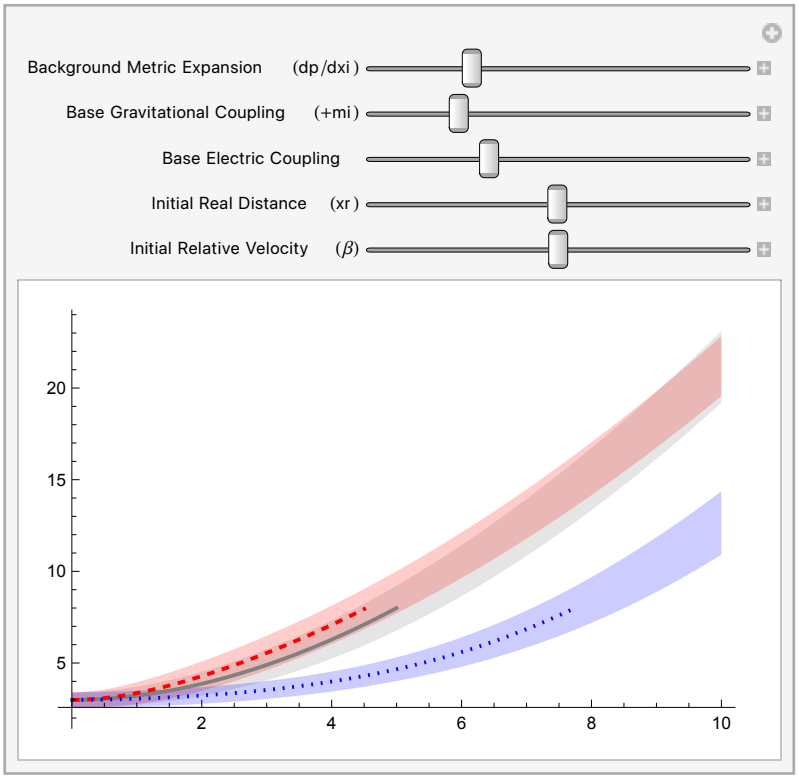
In[•]:=
```
Manipulate[
 Module[{tMax = 10, beta, eqOfMotion, solNeutral, solLike, solOpposite, soiNeutral,
   soiLike, soiOpposite, x, xi, t}, (*Normalized velocity*)beta[x_, xi_] := x'[xi];
  (*Background metric expansion function*)
  metric[xi_] := metricPressure * Exp[-0.1 * xi] + 0.1;
  (*Equation of Motion*)eqOfMotion[chargeSign_] :=
   x''[xi] == metric[xi] - (gForce * (1 + beta[x, xi]^2) / x[xi]^2) +
     (chargeSign * eForce * (1 - beta[x, xi]^2) / x[xi]^2);
  (*Solvers
   (Same as before,using Quiet to suppress singularity warnings at xr→0)*)
  solNeutral = Quiet@NDSolveValue[
     {eqOfMotion[0], x[0] == initialXr, x'[0] == initialBeta}, x, {xi, 0, tMax}];
  solLike = Quiet@NDSolveValue[
     {eqOfMotion[1], x[0] == initialXr, x'[0] == initialBeta}, x, {xi, 0, tMax}];
  solOpposite =
   Quiet@NDSolveValue[{eqOfMotion[-1], x[0] == initialXr, x'[0] == initialBeta,
      WhenEvent[x[xi] ≤ 0.01, "StopIntegration"]}, x, {xi, 0, tMax}];
  (*Dynamic Sphere of Influence (SOI) Radius Function SOI grows
   with effective mass (1+beta^2) and shrinks against metric pressure*)
  soiRadius[sol_, xi_] := (gForce * (1 + (sol'[xi])^2)) / (1 + metric[xi] * sol[xi]);
  (*Plotting the Traces with Shaded Spheres of Influence*)
  Show[(*Shaded regions representing the expanding/shrinking SOI*)
   Plot[{{solNeutral[xi] - soiRadius[solNeutral, xi],
      solNeutral[xi] + soiRadius[solNeutral, xi]},
     {solLike[xi] - soiRadius[solLike, xi], solLike[xi] + soiRadius[solLike, xi]},
     {solOpposite[xi] - soiRadius[solOpposite, xi],
      solOpposite[xi] + soiRadius[solOpposite, xi]}}, {xi, 0, tMax},
    Filling → {1 → {2}, 3 → {4}, 5 → {6}}, PlotStyle → None, FillingStyle →
     {1 → Directive[Gray, Opacity[0.2]], 3 → Directive[Red, Opacity[0.2]],
      5 → Directive[Blue, Opacity[0.2]]}], (*The hard geometric traces*)
   Plot[{solNeutral[xi], solLike[xi], solOpposite[xi]}, {xi, 0, tMax},
    PlotStyle → {Directive[Thick, Gray], Directive[Thick, Red, Dashed],
      Directive[Thick, Blue, Dotted]}, PlotRange → {{0, tMax}, {0, initialXr + 5}},
    AxesOrigin → {0, 0}, Frame → True, FrameLabel →
     {"Imaginary Axis / Metric Evolution (xi)", "Real Spatial Distance (xr)"}]]],
 (*Sliders*){{metricPressure, 0.5, "Background Metric Expansion (dp/dxi)"},
  0, 2},
 {{gForce, 1.0, "Base Gravitational Coupling (+mi)"}, 0.1, 5},
 {{eForce, 3.0, "Base Electric Coupling"}, 0, 10},
 {{initialXr, 3.0, "Initial Real Distance (xr)"}, 1.0, 5},
 {{initialBeta, 0.0, "Initial Relative Velocity (β)"}, -0.9, 0.9}]
```

Out[ ]=


Background Metric Expansion (dp/dxi)
Base Gravitational Coupling (+mi)
Base Electric Coupling
Initial Real Distance (xr)
Initial Relative Velocity (β)
20
15
10
5
2
4
6
8
10